

Hankel Determinants of Geometric Moments and Rational Lambert Values

Erdős Problem #1049

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ABSTRACT

We determine the fixed-base asymptotic of Zudilin's normalised Hankel determinants, including the constant and the factor $N^{-8F(1/q)}$, where $F(t) = \sum_{n \geq 1} (t^n - 1)^{-1}$. The proof uses positive geometric moments and a bound on the partition sum independent of N . Separately, we extend the rational-base irrationality region for F by calculating the degrees after cancellation in Zudilin's 2004 forms. The region includes every positive integral power of $31/4$, but excludes $3/2$.

1 Introduction

For each fixed real $q \in (0, 1)$, we determine the asymptotic size of the Hankel determinants arising from Zudilin's q -logarithm approximations. Write $(z; q)_m = \prod_{j=0}^{m-1} (1 - zq^j)$ and $F(t) = \sum_{n \geq 1} (t^n - 1)^{-1}$ for $t > 1$. At the auxiliary parameters $x = z = 1$, the normalisation in [5, (6), §4] gives

$$v_m^*(q) = \sum_{t \geq 0} q^{(m+1)t} \frac{(q; q)_m^3 (q^{t+1}; q)_m}{(q^{m+1+t}; q)_{m+1}}, \quad V_N^*(q) = \det(v_{i+j}^*(q))_{0 \leq i, j < N}. \quad (1)$$

The sums converge for $0 < q < 1$. They also define formal series, since the t th summand has q -order $(m+1)t$.

Theorem 1.1 (the size of V_N^* at a fixed base). *Fix $0 < q < 1$ and write $P = (q; q)_\infty$, $B_N = N(N-1)(2N-1)/6$ and $C_N = (N!)^2(N+1)!/2^N$. There is a $K(q) > 0$ with*

$$V_N^*(q) \sim K(q) C_N q^{B_N} P^{2N} N^{-8F(1/q)} \quad (N \rightarrow \infty).$$

The constant $K(q)$ is the convergent product in (13). Zudilin proved $\text{ord}_q V_N^* \geq B_N$ and the estimate $|V_N^*| \leq q^{N^3/3} \exp(O_q(N^2))$ [5, Lemma 1 and §4]. Theorem 1.1 determines the factorial, exponential and power factors contained in this estimate. We also recover the exact first formal term $C_N q^{B_N}$. These are different limiting questions:

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

multiplying by $(1 - q)^{N^3}$ preserves that first term and changes the fixed-base logarithm by a cubic quantity. No uniformity as $q \rightarrow 1$ is asserted.

We represent v_m^* as $\sum_{k \geq 0} a_k q^{(m+1)k}$ with $a_k > 0$. These are the moments of a positive measure supported on $1, q, q^2, \dots$. Heine's identity then expresses the determinant as a sum over choices of N nodes, with a squared Vandermonde factor for each choice [7, §2, (2)–(5)]. The contribution of the first N nodes has size $q^{B_N P^{2N}} \prod_{k < N} a_k$ up to a positive limiting factor. Moving the last node one place still gives a contribution of comparable size, so we must sum over displaced tuples. We index them by partitions. For each fixed partition, the weight ratios tend to 1; a summable bound independent of N then permits passage to the limit in the sum. The resulting limit is the same as for constant weights, for which Cauchy's determinant formula gives an exact evaluation.

The remaining calculation concerns the weights themselves. Their factorisation involves a two-fold and a three-fold convolution of the sequence $1/(q; q)_k$. Their first relative corrections are $-2F(1/q)/(k+1)$ and $-6F(1/q)/(k+2)$. Adding these corrections and then multiplying the weights over $k < N$ gives the power $N^{-8F(1/q)}$. Theorem 2.1 isolates the determinant argument from this coefficient calculation; it also evaluates the determinant with entries $(1 - q^{i+j+1})^{-2}$.

Chowla's conjecture, recorded by Erdős [1, p. 102], asks whether $F(t)$ is irrational for every rational $t > 1$. This is a related arithmetic question, and the fixed-base asymptotic alone does not answer it. In Section 4 we use the different 2004 construction [4, §5] to prove irrationality for coprime $a > b \geq 1$ satisfying

$$\frac{\log b}{\log a} < \theta^* = 0.4056830213840605 \dots,$$

where (15) defines θ^* exactly. The calculation cancels the common polynomial factors before clearing denominators at a/b . It includes every positive integral power of $31/4$. Neither this region nor the determinant argument settles the case $3/2$.

Section 2 proves the general determinant formula. Section 3 constructs the weights and proves Theorem 1.1. The rational-base argument is self-contained in Section 4, with longer calculations and complementary questions linked in Section 5. All logarithms are natural, empty products and determinants equal 1, and constants in $O_q(\cdot)$ may depend on the fixed q .

2 Geometric moment determinants

For constant weights, the moments are geometric series: $M_m = \sum_{k \geq 0} q^{(m+1)k} = (1 - q^{m+1})^{-1}$. Cauchy's determinant formula evaluates their determinant exactly. Writing

$$P = (q; q)_\infty, \quad M(q) = \prod_{d \geq 1} (1 - q^d)^{-d}, \quad B_N = \sum_{j < N} j^2,$$

we obtain

$$D_N^{(0)} := \det \left(\frac{1}{1 - q^{i+j+1}} \right)_{0 \leq i, j < N} = \frac{q^{B_N} \Delta_N}{\prod_{0 \leq i, j < N} (1 - q^{i+j+1})}, \quad \Delta_N = \prod_{d=1}^{N-1} (1 - q^d)^{2(N-d)}. \quad (2)$$

For each fixed d , the multiplicity of $1 - q^d$ in the denominator increases to d , giving the factor $M(q)$ in the limit. The numerator supplies two more copies:

$$\frac{\Delta_N}{p^{2N}} = \prod_{d < N} (1 - q^d)^{-2d} \prod_{d \geq N} (1 - q^d)^{-2N} \rightarrow M(q)^2.$$

Indeed, $\sum d |\log(1 - q^d)| < \infty$, and the logarithm of the second product is $O_q(Nq^N)$. Thus

$$D_N^{(0)} \sim M(q)^3 q^{B_N} p^{2N}. \quad (3)$$

Under the following hypotheses, replacing the constant weights by a_k multiplies the asymptotic by $\prod_{k < N} a_k$. The ratio limit controls each fixed displacement of a node, while the polynomial bound controls the sum over all displacements. The power weights $a_k = (k + 1)^s$, with $s \in \mathbb{R}$, satisfy both conditions.

Theorem 2.1 (geometric moment determinants). *Let $0 < q < 1$ and $a_k > 0$ for $k \geq 0$. Suppose that, for fixed constants $C > 0$ and $\kappa \geq 0$,*

$$\frac{a_{k+h}}{a_k} \rightarrow 1 \quad (k \rightarrow \infty) \text{ for each fixed } h \geq 0, \quad \frac{a_{k+h}}{a_k} \leq C(1+h)^\kappa \quad (k, h \geq 0). \quad (4)$$

For $M_m = \sum_{k \geq 0} a_k q^{(m+1)k}$ and $D_N = \det(M_{i+j})_{0 \leq i, j < N}$, we have

$$D_N \sim M(q)^3 q^{B_N} p^{2N} \prod_{k=0}^{N-1} a_k \quad (N \rightarrow \infty). \quad (5)$$

For every real s , the weights $a_k = (k + 1)^s$ satisfy (4) with $C = 1$ and $\kappa = \max(s, 0)$, and their first N terms have product $(N!)^s$. Taking $s = 1$ and summing the moments gives

$$\det \left(\frac{1}{(1 - q^{i+j+1})^2} \right)_{0 \leq i, j < N} \sim M(q)^3 N! q^{B_N} p^{2N}. \quad (6)$$

Proof of Theorem 2.1. The bound $a_h \leq Ca_0(1+h)^\kappa$ ensures convergence of every moment. Apply Cauchy–Binet to a finite set of nodes and let its size tend to infinity. The matrix entries converge, while positivity permits passage to the limit in the sum, giving

$$D_N = \sum_{0 \leq k_0 < \dots < k_{N-1}} \prod_{i < N} a_{k_i} q^{k_i} \prod_{i < j < N} (q^{k_i} - q^{k_j})^2. \quad (7)$$

The tuple $k_i = i$ contributes $q^{B_N} \Delta_N \prod_{i < N} a_i$. It does not account for the whole asymptotic. For example, replace $(0, 1, \dots, N-2, N-1)$ by $(0, 1, \dots, N-2, N)$. The ratio of its contribution to that of the original tuple is

$$q \frac{a_N}{a_{N-1}} \prod_{d=1}^{N-1} \left(\frac{1 - q^{d+1}}{1 - q^d} \right)^2 = q \frac{a_N}{a_{N-1}} \left(\frac{1 - q^N}{1 - q} \right)^2 \rightarrow \frac{q}{(1 - q)^2}. \quad (8)$$

Only pairs involving the moved node change, so the product telescopes. At $q = 1/2$ the limiting ratio is 2: the displaced tuple contributes twice as much asymptotically as $(0, 1, \dots, N-1)$.

Subtracting $(0, 1, \dots, N-1)$ gives the shifts $\lambda_i = k_i - i$. Since $k_{i+1} \geq k_i + 1$, these are nonnegative and weakly increasing; reversing their order gives a partition. For $N = 4$, for example,

$$\frac{(0, 1, 3, 5)}{(k_i)} \mapsto \frac{(0, 0, 1, 2)}{(\lambda_i)} \mapsto \frac{(2, 1, 0, 0)}{(\mu_j)}.$$

In general, put $\mu_j = \lambda_{N-j}$ for $1 \leq j \leq N$, so that $\mu_1 \geq \dots \geq \mu_N \geq 0$. Division by the contribution of $k_i = i$ turns the summand for this tuple into

$$W_N(\mu) = q^{\sum_{j \leq N} (2j-1)\mu_j} \prod_{j \leq N} \frac{a_{N-j+\mu_j}}{a_{N-j}} \cdot \prod_{1 \leq j < k \leq N} \left(\frac{1 - q^{k-j+\mu_j-\mu_k}}{1 - q^{k-j}} \right)^2. \quad (9)$$

The j th index from the right contributes once through q^{k_i} and twice through each of its $j-1$ pairs with later indices in the squared Vandermonde. Its displacement therefore has exponent $1 + 2(j-1) = 2j-1$.

Set $W_N(\mu) = 0$ when μ has more than N positive parts, so that every sum has the same index set. Fix a partition with ℓ positive parts. It affects only the last ℓ indices of $(0, 1, \dots, N-1)$: for instance, $(2, 1)$ affects the last two for every $N \geq 2$. Those indices tend to infinity, so the weight ratios in (9) tend to 1 by the fixed-shift hypothesis.

The Vandermonde quotient still contains a growing number of factors, but its dependence on N telescopes. The factors with $j < k \leq \ell$ are fixed, and those with $j > \ell$ equal 1. For $N \geq \ell$ and each $j \leq \ell$, the remaining factors satisfy

$$\prod_{k=\ell+1}^N \frac{1 - q^{k-j+\mu_j}}{1 - q^{k-j}} = \frac{(q^{N+1-j}; q)_{\mu_j}}{(q^{\ell+1-j}; q)_{\mu_j}} \rightarrow \frac{1}{(q^{\ell+1-j}; q)_{\mu_j}}. \quad (10)$$

Thus each $W_N(\mu)$ has the same limit as for constant weights.

To pass this limit through the sum over partitions, we need a bound independent of N that remains summable as both the number and size of the parts grow. Only factors with $j \leq \ell$ can differ from 1. For each such j , the denominator product in the Vandermonde quotient is at least P , and its numerator product is at most 1. Together with the polynomial shift bound, this gives

$$W_N(\mu) \leq (CP^{-2})^\ell \prod_{j=1}^{\ell} (1 + \mu_j)^\kappa q^{(2j-1)\mu_j}.$$

Let $A = \sum_{u \geq 1} (1+u)^\kappa q^{u-1} < \infty$. Dropping the ordering of the positive parts, we obtain

$$\begin{aligned} \sum_{\ell(\mu)=\ell} \sup_N W_N(\mu) &\leq (CP^{-2})^\ell \prod_{j=1}^{\ell} \sum_{u \geq 1} (1+u)^\kappa q^{(2j-1)u} \\ &\leq (CP^{-2}A)^\ell q^{\ell^2}. \end{aligned}$$

Here $q^{(2j-1)u} \leq q^{2j-1}q^{u-1}$ and $1 + 3 + \dots + (2\ell - 1) = \ell^2$. Since $q < 1$, the bound $(CP^{-2}A)^\ell q^{\ell^2}$ is summable over ℓ , independently of N .

Dominated convergence now shows that $\sum_\mu W_N(\mu)$ has the same limit as for $a_k = 1$. This common limit is positive, since the empty partition contributes 1. Restoring the contribution of $(0, 1, \dots, N-1)$ gives

$$\frac{D_N}{D_N^{(0)} \prod_{k < N} a_k} \rightarrow 1.$$

Combining this limit with (3) proves the theorem. $\square$

3 The weights of Zudilin's moments

We apply Theorem 2.1 by collecting the dependence on m in (1) into the single variable $w = q^{m+1}$. For this purpose, set

$$G_q(w) = \frac{1}{(w; q)_\infty^3} \sum_{t \geq 0} \frac{w^t}{(q; q)_t} \frac{(q^t w^2; q)_\infty}{(q^t w; q)_\infty^2}, \quad a_k(q) = [w^k] P^4 G_q(w). \quad (11)$$

The product identities $(q; q)_m = P / (q^{m+1}; q)_\infty$ and $(q^a; q)_m = (q^a; q)_\infty / (q^{a+m}; q)_\infty$ give

$$v_m^* = P^4 G_q(q^{m+1}) = \sum_{k \geq 0} a_k(q) q^{(m+1)k}. \quad (12)$$

These identities hold formally in $\mathbb{Z}[[q, w]]$ and, for fixed $0 < q < 1$, analytically for $|w| < 1$. In the formal identity only finitely many t contribute to any w -coefficient. For the analytic identity, we use the absence of denominator zeros in that disc and the locally uniform bounds on the product tails. The factor w^t then gives locally uniform convergence of the sum. To prove positivity and obtain a bound uniform in the shift, we express its coefficients as finite sums.

3.1 Positive weights

For $r \geq 1$ and $k \geq 0$, set

$$R_k^{(r)}(q) = \sum_{n_1 + \dots + n_r = k} \frac{(q; q)_k}{\prod_{j=1}^r (q; q)_{n_j}},$$

over r nonnegative parts. These are the multivariate Rogers–Szegő polynomials at unit arguments, written as sums of Gaussian multinomial polynomials [19]. At $q = 0$ they count compositions, and

$$c_k := R_k^{(2)}(0) R_k^{(3)}(0) = \frac{(k+1)^2(k+2)}{2}, \quad \prod_{k < N} c_k = C_N.$$

Proposition 3.1 (the positive moment weights). *For every $k \geq 0$,*

$$a_k = P^4 \frac{R_k^{(2)} R_k^{(3)}}{(q; q)_k}.$$

At fixed $0 < q < 1$ these weights satisfy

$$P^5 c_k \leq a_k \leq P^{-1} c_k, \quad \frac{a_{k+h}}{a_k} \leq P^{-6} (1+h)^3 \quad (k, h \geq 0).$$

Proof. We use the Gaussian binomial coefficients $\begin{bmatrix} n \\ j \end{bmatrix}_q = (q; q)_n / ((q; q)_j (q; q)_{n-j})$ for $0 \leq j \leq n$, and put $\mathcal{E}(z) = (z; q)_\infty^{-1}$. Euler's expansion gives $\mathcal{E}(z)^r = \sum_{n \geq 0} R_n^{(r)} z^n / (q; q)_n$. To obtain $R_k^{(2)} R_k^{(3)}$, we shift the coefficients of $\mathcal{E}^3$ and sum over decompositions of the index. The unnormalised q -difference operator $\partial_q f(z) = (f(z) - f(qz))/z$ is convenient because $\partial_q z^j = (1 - q^j) z^{j-1}$ cancels the last factor of $(q; q)_j$. In particular,

$$\partial_q \mathcal{E} = \mathcal{E}, \quad \partial_q^n \mathcal{E}(z)^3 = \sum_{j \geq 0} \frac{R_{n+j}^{(3)}}{(q; q)_j} z^j.$$

The q -Leibniz rule, obtained by induction from the product rule, is

$$\partial_q^n (fg)(z) = \sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix}_q (\partial_q^j f)(q^{n-j} z) (\partial_q^{n-j} g)(z).$$

Let $h_k(z) = \sum_{j=0}^k \begin{bmatrix} k \\ j \end{bmatrix}_q (z; q)_j$. Applying the rule to $\mathcal{E} \cdot \mathcal{E}$ and using $\mathcal{E}(q^m z) = (z; q)_m \mathcal{E}(z)$, we obtain $\partial_q^k \mathcal{E}(z)^2 = \mathcal{E}(z)^2 h_k(z)$ (for $k = 1$, both sides equal $\mathcal{E}(z)^2(2 - z)$). A second application, this time to $\mathcal{E} \cdot \mathcal{E}^2$, gives

$$\partial_q^n \mathcal{E}(z)^3 = \mathcal{E}(z)^3 \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (z; q)_k h_k(z),$$

and hence

$$\sum_{n \geq 0} \frac{w^n}{(q; q)_n} \partial_q^n \mathcal{E}(z)^3 = \mathcal{E}(w) \mathcal{E}(z)^3 \sum_{k \geq 0} \frac{w^k (z; q)_k}{(q; q)_k} h_k(z).$$

The left-hand side is $\sum_{n, j \geq 0} R_{n+j}^{(3)} w^n z^j / ((q; q)_n (q; q)_j)$. Putting $z = w$ collects all pairs with the same sum $n + j$. The coefficient of w^k is consequently $R_k^{(3)} \sum_{j \leq k} ((q; q)_j (q; q)_{k-j})^{-1} = R_k^{(2)} R_k^{(3)} / (q; q)_k$. On the right-hand side, put $z = w$, expand h_k , and write $k = j + t$. The identity $(w; q)_{j+t} = (w; q)_j (wq^j; q)_t$ gives

$$\mathcal{E}(w)^4 \sum_{j \geq 0} \frac{w^j (w; q)_j^2}{(q; q)_j} \sum_{t \geq 0} \frac{(wq^j; q)_t}{(q; q)_t} w^t.$$

The inner sum is evaluated by the q -binomial theorem:

$$\sum_{t \geq 0} \frac{(wq^j; q)_t}{(q; q)_t} w^t = \frac{(w^2 q^j; q)_\infty}{(w; q)_\infty}.$$

Finally, $(w; q)_j = (w; q)_\infty / (wq^j; q)_\infty$ cancels two of the five factors $\mathcal{E}(w)$, leaving

$$\mathcal{E}(w)^3 \sum_{j \geq 0} \frac{w^j}{(q; q)_j} \frac{(w^2 q^j; q)_\infty}{(wq^j; q)_\infty^2} = G_q(w).$$

We may read each identity coefficientwise, or as an absolutely convergent expansion when $|z|$ and $|w|$ are sufficiently small.

For the bounds, we write $b_k^{(r)} = [z^k](z; q)_\infty^{-r}$, so that $R_k^{(r)} = (q; q)_k b_k^{(r)}$ and $a_k = P^4(q; q)_k b_k^{(2)} b_k^{(3)}$. By Euler's expansion, $b_k^{(r)}$ is an r -fold convolution of $1/(q; q)_n$, whose terms lie in $[1, P^{-1}]$. There are $\binom{k+r-1}{r-1}$ compositions of k into r nonnegative parts, and we deduce

$$\binom{k+r-1}{r-1} \leq b_k^{(r)} \leq P^{-r} \binom{k+r-1}{r-1}.$$

Combining these at $r = 2, 3$ with $P \leq (q; q)_k \leq 1$ and $c_k = (k+1) \binom{k+2}{2}$ gives $P^5 c_k \leq a_k \leq P^{-1} c_k$. The shift bound follows from $c_{k+h}/c_k \leq (1+h)^3$. $\square$

It remains to prove the fixed-shift limit and determine the product $\prod_{k < N} a_k$. Both follow from the first-order asymptotic of a_k/c_k .

3.2 The coefficient asymptotic

Proof of Theorem 1.1. Put $L = F(1/q)$, $e_k = (q; q)_k^{-1}$ and $E = P^{-1}$. Replacing every e_k by E in the two convolution sums gives $E^2(k+1)$ and $E^3 \binom{k+2}{2}$. The factors $k+1$ and $\binom{k+2}{2}$ count nonnegative pairs and triples with sum k . For a fixed i , one pair and $k-i+1$ triples have first coordinate i . Consequently, the summable differences $E - e_i$ produce a constant correction in the two-fold convolution and a linear correction in the three-fold convolution. The terms containing two or more such differences contribute to the error bounds below.

Writing $d_k = E - e_k$, the identity $e_{k+1} - e_k = q^{k+1} e_{k+1}$ gives

$$d_k = \sum_{j > k} q^j e_j, \quad 0 \leq d_k \leq \frac{E q^{k+1}}{1-q}.$$

Thus $\sum d_k$ and $\sum k d_k$ converge. To evaluate the former, use Euler's identity $\mathcal{E}(z) = \sum_{k \geq 0} e_k z^k = (z; q)_\infty^{-1}$ for $|z| < 1$. Tonelli's theorem and logarithmic differentiation at $z = q$ give

$$S := \sum_{k \geq 0} d_k = \sum_{j \geq 1} j q^j e_j = q \mathcal{E}'(q) = E \sum_{r \geq 1} \frac{q^r}{1-q^r} = EL.$$

The differentiation is justified by locally uniform convergence on $|z| < 1$.

Let $b_k^{(r)} = [z^k] \mathcal{E}(z)^r$. The finite-sum formula in Proposition 3.1 becomes

$$a_k = P^4 (q; q)_k b_k^{(2)} b_k^{(3)}.$$

We expand each finite convolution using $e_k = E - d_k$. For two factors,

$$b_k^{(2)} = E^2(k+1) - 2E \sum_{i=0}^k d_i + \sum_{i=0}^k d_i d_{k-i} = E^2(k+1 - 2L) + O_q((k+1)q^k).$$

For three factors, the terms with one d_i contribute $-3E^2 \sum_{i=0}^k (k-i+1) d_i$. Here the sum equals $(k+1)S + O_q(1)$ because $\sum i d_i < \infty$. Terms with two d_i are bounded by $3ES^2$, and the term with three is $O_q((k+1)^2 q^k)$. It follows that

$$b_k^{(3)} = E^3 \left(\binom{k+2}{2} - 3L(k+1) \right) + O_q(1).$$

Dividing the correction $-3L(k+1)$ by the leading term $\binom{k+2}{2}$ gives the relative correction $-6L/(k+2)$. The two-fold convolution contributes $-2L/(k+1)$ in the same way. Since $(q; q)_k = P(1 + O_q(q^k))$, the leading constants in $a_k = P^4(q; q)_k b_k^{(2)} b_k^{(3)}$ multiply to $P^5 E^5 = 1$. With $c_k = (k+1)^2(k+2)/2$, the two relative corrections therefore give

$$\begin{aligned} \frac{a_k}{c_k} &= \left(1 - \frac{2L}{k+1} + O_q(q^k)\right) \left(1 - \frac{6L}{k+2} + O_q((k+1)^{-2})\right) (1 + O_q(q^k)) \\ &= 1 - \frac{8L}{k+1} + O_q((k+1)^{-2}). \end{aligned}$$

In particular $a_{k+1}/a_k \rightarrow 1$, and multiplication of finitely many adjacent ratios gives $a_{k+h}/a_k \rightarrow 1$ for each fixed h . The uniform shift bound follows from Proposition 3.1, so Theorem 2.1 applies.

Let γ_E denote Euler's constant and put

$$A(q) = e^{-8\gamma_E L} \prod_{k \geq 0} \left(\frac{a_k}{c_k} e^{8L/(k+1)} \right), \quad K(q) = A(q) M(q)^3. \quad (13)$$

Positivity of a_k and summability of $\log(a_k/c_k) + 8L/(k+1) = O_q((k+1)^{-2})$ show that $A(q)$ converges to a positive number. Since $\prod_{k < N} c_k = C_N$, we can separate these contributions exactly:

$$\log \frac{\prod_{k < N} a_k}{C_N} = -8L \sum_{k < N} \frac{1}{k+1} + \sum_{k < N} \left(\log \frac{a_k}{c_k} + \frac{8L}{k+1} \right).$$

The second sum tends to $\log A(q) + 8\gamma_E L$. Using $\sum_{k < N} (k+1)^{-1} = \log N + \gamma_E + o(1)$, the Euler-constant terms cancel, giving $\prod_{k < N} a_k \sim A(q) C_N N^{-8L}$. Substitution in (5) proves the theorem, or equivalently

$$\log V_N^* = B_N \log q + \log C_N + 2N \log P - 8L \log N + \log K(q) + o(1).$$

□

Thus the product of the cubic weights c_k gives C_N , whereas the first correction to a_k/c_k determines $N^{-8F(1/q)}$.

3.3 The first nonzero formal term

The order of a nonzero formal series is the least exponent with nonzero coefficient. We now fix N and determine the first term in the formal q -expansion. The shifted term in (8) then has an extra factor of q . More generally, every nonempty partition has positive displacement exponent. Thus a single tuple determines the first formal term, although all partitions were needed for the fixed-base constant.

Theorem 3.2 (the first nonzero term of the Hankel determinant). *For every $N \geq 1$,*

$$\text{ord}_q V_N^* = \frac{N(N-1)(2N-1)}{6},$$

and the coefficient of the first nonzero monomial is

$$[q^{N(N-1)(2N-1)/6}] V_N^* = \frac{(N!)^2 (N+1)!}{2^N}.$$

Proof. At $q = 0$, the $t = 0$ term of $G_q(w)$ is $(1 - w^2)/(1 - w)^5$, while the remaining terms sum to $w/(1 - w)^4$. Thus

$$G_0(w) = \frac{1 + 2w}{(1 - w)^4}, \quad a_k(0) = c_k > 0.$$

We compute modulo q^{M+1} by truncating each moment sum in (12) at $k = M$. Since all omitted terms are divisible by q^{M+1} , multilinearity leaves the determinant unchanged modulo that power. Applying finite Cauchy–Binet and then allowing M to increase, we obtain the coefficientwise identity

$$V_N^* = \sum_{k_0 < \dots < k_{N-1}} \left(\prod_{i < N} a_{k_i}(q) q^{k_i} \right) \prod_{i < j < N} (q^{k_i} - q^{k_j})^2.$$

The order of the summand indexed by (k_i) is $\sum_{i < N} (2N - 1 - 2i)k_i$. Because $k_i \geq i$ and all the weights $2N - 1 - 2i$ are positive, we obtain the unique minimum at $k_i = i$, with order and coefficient

$$\sum_{i < N} (2N - 1 - 2i)i = B_N, \quad \prod_{i < N} c_i = C_N.$$

No other tuple can cancel that coefficient. For $N = 2$, the pair $(0, 1)$ contributes $6q + O(q^2)$, and every other increasing pair has order at least two. $\square$

This proves equality in Zudilin’s bound [5, §4, pp. 6–7]. An alternative argument, which computes the first nonzero term of every transformed row, is given in [the companion paper](#).

4 Rational bases

We now use the 2004 coefficient family. If $U, V \in \mathbb{Z}[X]$ have degree at most W , evaluation at a/b requires the factor b^W to produce integer coefficients. A common polynomial factor should therefore be cancelled first. For example, at $3/2$ the pair $(X + 1)(X + 2), (X + 1)(X + 3)$ gives $(35, 45)$ after clearing to degree 2. After cancelling $X + 1$, clearing to degree 1 gives $(7, 9)$. For Zudilin’s forms, the degrees are of quadratic order in the index n . Write K_n for the degree of the original coefficient and W_n for its degree after cancellation. The cancelled remainder has leading size $x^{-(K_n - W_n)}$ at a fixed real base $x > 1$. At $x = a/b$, clearing the coefficients therefore has the balance

$$b^{W_n} (a/b)^{-(K_n - W_n)} = \frac{b^{K_n}}{a^{K_n - W_n}}. \quad (14)$$

The proof below shows that the remaining factors contribute only $o(n^2)$ to the logarithm. Hence the limits $K_n/n^2 \rightarrow C_1$ and $(K_n - W_n)/n^2 \rightarrow C_0$ give the sufficient inequality $C_1 \log b < C_0 \log a$. We first justify cancellation in $\mathbb{Z}[X]$ and calculate the degrees, before specialising X to a/b .

We use the parameter direction from Zudilin’s construction [4, §5, pp. 161–162], together with its thirteen intervals and constants C_1, C_0 . At integer bases his theorem

gives the irrationality-exponent bound $C_1/C_0 = 2.46497868\dots$ [4, Thm. 1, p. 154]. Let $\psi_1(u) = \sum_{k \geq 0} (k+u)^{-2}$ for $u > 0$, and let $\mathcal{I}$ consist of the thirteen intervals listed in the proof. We put

$$C_1 = \frac{1091}{2}, \quad J = \sum_{[u,v] \in \mathcal{I}} (\psi_1(u) - \psi_1(v)), \quad C_0 = 266 - \frac{3}{\pi^2}(225 - J). \quad (15)$$

The intervals are disjoint and lie in $[1/14, 1)$, so $0 \leq J \leq \psi_1(1/14) - \psi_1(1) < 196$. Together with $\pi > 3$, these bounds give $0 < C_0 < 266 < C_1/2$. Thus $\theta^* = C_0/C_1$ and $\mu = C_1/C_0$ are positive reciprocal constants. The notation $\mu_{\text{irr}}(\xi)$ instead denotes the irrationality exponent of a value. The estimates below give a sufficient cutoff; its optimality is unknown.

Theorem 4.1 (rational-base region for Zudilin's forms). *Let $a > b \geq 1$ be coprime integers with*

$$b^\mu < a, \quad \text{equivalently} \quad \frac{\log b}{\log a} < \theta^*,$$

$$\theta^* = \frac{C_0}{C_1} = 0.4056830213840605\dots,$$

$$\mu = \frac{C_1}{C_0} = 2.4649786835749750\dots$$

Then $F(a/b)$ is irrational.

The condition holds for every integer base, since $b = 1$ gives $\log b / \log a = 0$. For a fixed denominator $b > 1$, it requires $a > b^{2.46497868\dots}$, much more than $a > b$; nevertheless it gives infinitely many reduced noninteger rational bases for each such b . The base $31/4$ satisfies the condition, and taking a common positive integer power of the numerator and denominator leaves their logarithmic ratio unchanged. Since $\theta^* < 1/2 < \log 2 / \log 3$, the condition fails at $3/2$. At equality $\log b / \log a = \theta^*$ the quadratic exponent vanishes, so the estimates below give no conclusion.

Proof of Theorem 4.1. We first identify the integral polynomials obtained by cancellation. We then compute their degrees and the size of the cleared remainder, which will justify (14) to quadratic order.

The source forms and polynomial cancellation. Fix $n \geq 1$ and set

$$a_0 = 14n + 1, \quad a_1 = 12n + 1, \quad a_2 = 14n + 1, \quad \beta = 27n + 2, \quad N = 15n.$$

In this proof N denotes the cyclotomic cutoff. We use the coefficient pair A_n, B_n of the source identities [4, (8)–(11), pp. 156–157], with $H_n = A_n F - B_n$. Let Φ_ℓ be the cyclotomic polynomial whose roots are the primitive ℓ th roots of unity. Put $D_N(X) = \prod_{\ell=1}^N \Phi_\ell(X)$, $M_n = 266n^2 + 34n + 1$, and

$$\Omega_n(X) = \prod_{\ell=2}^N \Phi_\ell(X)^{\nu_\ell}, \quad \nu_\ell = \omega(n/\ell),$$

where the periodic function

$$\omega(x) = \max\{0, \lfloor 14x \rfloor + \lfloor 13x \rfloor - \lfloor 12x \rfloor - \lfloor 15x \rfloor, \\ 2\lfloor 14x \rfloor - \lfloor 13x \rfloor - \lfloor 15x \rfloor\}$$

has period 1: the coefficients of the floor arguments in each difference have equal sums, $14 + 13 = 12 + 15$ and $2 \cdot 14 = 13 + 15$. On $[0, 1)$ it is zero or one, with support

$$\begin{aligned} &[1/14, 1/12), [1/7, 1/6), [3/14, 1/4), [2/7, 1/3), \\ &[5/14, 2/5), [3/7, 7/15), [1/2, 8/15), [4/7, 3/5), \\ &[9/14, 2/3), [5/7, 11/15), [11/14, 4/5), \\ &[6/7, 13/15), [13/14, 14/15). \end{aligned}$$

These are the intervals $\mathcal{I}$ used to define J . The function ω , these thirteen intervals and the exponents $\nu_\ell = \omega(n/\ell)$ of the source's (22) are printed at [4, pp. 161–162].

We first bound the remainder before cancellation. For $x > 1$ we put $q = 1/x$ and use identity (9) on p. 156 of the source, which gives the positive representation

$$H_n(x) = \sum_{t \geq 0} q^{a_0 t} \frac{(q^{t+1}; q)_{a_1-1}}{(q; q)_{a_1-1}} \frac{(q; q)_{\beta-a_2-1}}{(q^{a_2+t}; q)_{\beta-a_2}}.$$

The denominator length is $\beta - a_2$, as in the gamma expression (7) and residues (8) of [4, p. 156]; the unnumbered display of $R(T)$ on that page has length $\beta - a_2 - 1$ instead. Each of the four finite products displayed above lies between $P = (q; q)_\infty > 0$ and 1, so

$$P^2 \leq H_n(x) \leq \frac{P^{-2}}{1 - q^{a_0}}.$$

In particular $H_n(x) > 0$ and $\log H_n(x) = O_x(1)$.

To identify the integral polynomials we use the inclusion in Zudilin's Lemma 7, display (23) [4, p. 161]. The proof compares the cyclotomic valuations of the coefficient polynomials under six admissible parameter permutations, and gives the divisibility in $\mathbb{Z}[X]$ that we need before evaluation. Its parameter vector is $n(13, 14, 12, 14, 15, 13)$, its maximum is $15n$, and $\beta - a_1 - a_2 = n > 0$, which is the positivity condition $s > 0$ of the lemma. The conditions (14) of [4, p. 157], $a_1 \leq a_2$ and $a_1 + a_2 \leq \beta \leq a_0 + a_2$, hold as well, so the exponent in (23) is the integer (16), which equals M_n . Lemma 7 therefore gives, for every $n \geq 1$,

$$\Lambda_n(X) = X^{-M_n} \frac{D_N(X)}{\Omega_n(X)} H_n(X) = U_n(X)F(X) - V_n(X), \quad U_n, V_n \in \mathbb{Z}[X].$$

This is the polynomial conclusion before integer specialisation in the source's (24), p. 162. The coefficients are the cancelled source coefficients:

$$U_n = X^{-M_n} (D_N / \Omega_n) A_n, \quad V_n = X^{-M_n} (D_N / \Omega_n) B_n.$$

Indeed, coefficients of F and 1 over $\mathbb{Q}(X)$ are unique because F is not a rational function. To see this, we split $F(e^h)$ at $m = 1/h$ as $h \downarrow 0$. For $m \leq 1/h$, the inequalities $y^{-1} - 1 \leq (e^y -$

$1)^{-1} \leq y^{-1}$ for $0 < y \leq 1$ give $h^{-1} \sum_{m \leq 1/h} m^{-1} + O(h^{-1})$. Bounding the remaining terms by a geometric tail gives a further $O(h^{-1})$, and therefore $F(e^h) = h^{-1} \log(1/h) + O(h^{-1})$, which has no rational-function pole order at $X = 1$. The cancelled remainder is positive for $x > 1$, since D_N/Ω_n is a product of cyclotomic polynomials positive there.

The degrees after cancellation. The Gaussian binomial polynomial $\begin{bmatrix} m \\ k \end{bmatrix}_X = \prod_{j=1}^k (1 - X^{m-k+j})/(1 - X^j)$ is monic of degree $k(m - k)$ for integers $0 \leq k \leq m$. The source's (8) and (10) on p. 156 give the coefficient whose degree we need:

$$A_n(X) = \sum_{k=a_2}^{\beta-1} (-1)^{a_1+a_2+k+1} X^{a_0k + \binom{a_1}{2} - \binom{\beta-a_2}{2} + \binom{\beta-k}{2}} \begin{bmatrix} k-1 \\ a_1-1 \end{bmatrix}_X \begin{bmatrix} \beta-a_2-1 \\ \beta-k-1 \end{bmatrix}_X.$$

Let $d_{n,k}$ be the degree of the k th summand. Adding the monomial exponent and the degrees of the two Gaussian factors gives

$$\begin{aligned} d_{n,k} &= a_0k + \binom{a_1}{2} - \binom{\beta-a_2}{2} + \binom{\beta-k}{2} \\ &\quad + (a_1-1)(k-a_1) + (\beta-k-1)(k-a_2) \\ &= \frac{-k^2 + 80kn + 3k - 340n^2 - 26n}{2}. \end{aligned}$$

Hence $d_{n,k+1} - d_{n,k} = 40n + 1 - k > 0$ for $a_2 \leq k \leq \beta - 2 = 27n$. The last summand alone therefore determines the leading term, giving

$$K_n := \deg A_n = \frac{1091n^2 + 81n + 2}{2}, \quad W_n := \deg U_n = K_n - M_n + \sum_{\ell \leq 15n} (1 - \nu_\ell) \varphi(\ell).$$

Here $\varphi(\ell) = \deg \Phi_\ell$ is Euler's totient function, and $\nu_1 = \omega(n) = 0$. In particular,

$$K_n - W_n = M_n - \deg(D_N/\Omega_n).$$

The factor X^{-M_n} lowers the degree by M_n , whereas the remaining cyclotomic factors increase it by $\deg(D_N/\Omega_n)$. Their difference will determine the decay of the cancelled remainder.

The bounds for H_n are uniform for $x \geq 2$: then $P \geq \prod_{j \geq 1} (1 - 2^{-j}) > 0$ and $(1 - x^{-a_0})^{-1} \leq 2$. For fixed n , therefore, $H_n(x) = O(1)$ and $\Lambda_n(x) = O(x^{W_n - K_n})$ as $x \rightarrow \infty$. The unique top summand of A_n has leading coefficient $(-1)^{a_1+a_2+\beta} = (-1)^n$. Since the Gaussian factors and D_N/Ω_n are monic, U_n has that leading coefficient too. Moreover $F(x) = x^{-1} + O(x^{-2})$ and $K_n \geq 2$, so

$$V_n(x) = U_n(x)F(x) - \Lambda_n(x) = (-1)^n x^{W_n-1} + O(x^{W_n-2}).$$

Here $W_n \geq K_n - M_n = (559n^2 + 13n)/2 > 0$. Thus $\deg V_n = W_n - 1$, with the same unit leading coefficient as U_n . For every prime $p \mid b$, reduction of the cleared first coordinate gives

$$b^{W_n} U_n(a/b) \equiv (-1)^n a^{W_n} \not\equiv 0 \pmod{p}.$$

Thus $U_n(a/b)$ has reduced denominator exactly b^{W_n} , the least common clearing denominator for this pair. Every common integer divisor of the cleared row is consequently coprime to b . This leaves common factors at other primes and savings from combinations of different rows to be analysed separately.

The quadratic degree limits. The degree calculation kept n fixed and let $x \rightarrow \infty$. For the irrationality argument we now keep x fixed and let $n \rightarrow \infty$. We first find the quadratic part of the degree removed by cancellation. The limits below are the cyclotomic limits of [4, Lemmas 1–2, p. 155]. The proof uses the argument through reciprocal intervals, the summatory totient estimate and the trigamma function from the proof of [3, Lemma 1, p. 466], with an explicit truncation of the block sum. The elementary summatory estimate $\sum_{\ell \leq y} \varphi(\ell) = 3y^2/\pi^2 + O(y \log y)$ gives

$$\frac{1}{n^2} \sum_{\ell \leq 15n} \varphi(\ell) \rightarrow \frac{675}{\pi^2}.$$

For $[u, v) \in \mathcal{I}$, the condition $\{n/\ell\} \in [u, v)$ is the disjoint union of intervals $n/(k+v) < \ell \leq n/(k+u)$ for $k \geq 0$. We may apply the same summatory estimate term by term to any finite set of k . If we stop before $k = K$, with $K \geq 1$, every omitted block has $\ell \leq n/(K+u)$. Thus the elementary bound $\sum_{\ell \leq y} \varphi(\ell) \leq y^2$ for $y \geq 0$ gives the uniform estimate

$$\frac{1}{n^2} \sum_{\ell \leq n/(K+u)} \varphi(\ell) \leq \frac{1}{(K+u)^2}.$$

Indeed, for $y \geq 1$ one can bound $\varphi(\ell)$ by ℓ and sum; for $0 \leq y < 1$ the sum is empty. This bound is uniform in n . We can now let n and then K tend to infinity, obtaining

$$\frac{1}{n^2} \sum_{\ell \leq 15n} \nu_\ell \varphi(\ell) \rightarrow \frac{3}{\pi^2} \sum_{[u,v) \in \mathcal{I}} \sum_{k \geq 0} \left(\frac{1}{(k+u)^2} - \frac{1}{(k+v)^2} \right) = \frac{3J}{\pi^2}.$$

Here no contributing index exceeds $14n$, since $u \geq 1/14$. Consequently

$$K_n/n^2 \rightarrow C_1, \quad (K_n - W_n)/n^2 \rightarrow C_0.$$

The cleared remainder. We return to a fixed real base $x > 1$ and estimate the cyclotomic product. Writing μ_{Mob} for the Möbius function, we have

$$\log \Phi_\ell(x) - \varphi(\ell) \log x = \sum_{d|\ell} \mu_{\text{Mob}}(d) \log(1 - x^{-\ell/d}).$$

The total absolute error over $\ell \leq 15n$ is $O_x(n)$, since it is at most

$$15n \sum_{d \geq 1} \frac{-\log(1 - x^{-d})}{d}.$$

The series converges because $-\log(1 - x^{-d}) \leq x^{-d}/(1 - x^{-1})$. Since $0 \leq 1 - \nu_\ell \leq 1$, it follows that

$$\log \Lambda_n(x) = -(K_n - W_n) \log x + O_x(n).$$

We can now clear the denominators at the rational base. Since $U_n, V_n \in \mathbb{Z}[X]$ and their degrees are at most W_n , the numbers $b^{W_n}U_n(a/b)$ and $b^{W_n}V_n(a/b)$ are integers. Moreover

$$\begin{aligned} \log(b^{W_n}\Lambda_n(a/b)) &= K_n \log b - (K_n - W_n) \log a + O_{a/b}(n) \\ &= (C_1 \log b - C_0 \log a)n^2 + o(n^2). \end{aligned}$$

The coefficient is negative under the theorem's hypothesis. The positive values of these integer-coefficient forms therefore tend to zero. If $F(a/b) = r/s$ with integers r, s and $s \neq 0$, every nonzero value would have absolute value at least $1/|s|$, a contradiction. $\square$

Corollary 4.2. $F((31/4)^r)$ is irrational for every integer $r \geq 1$.

Proof. The exact inequalities $31^2 < 4^5$ and $4^{200} < 31^{81}$ give $2/5 < \log 4 / \log 31 < 81/200$. The first term of each trigamma difference yields

$$J \geq \sum_{[u,v] \in \mathcal{I}} (u^{-2} - v^{-2}) = \frac{2015640690251}{25971865920}.$$

Using $\pi > 157/50$ gives the rational lower bound

$$\theta^* > \frac{2359630009523263}{5820307922172744} > \frac{81}{200}.$$

Taking a common positive integer power preserves both the logarithmic ratio and coprimality, so the same criterion applies for every $r \geq 1$. $\square$

Corollary 4.3 (an irrationality measure uniform over powers). *For coprime $a > b \geq 1$ with $\theta = \log b / \log a < \theta^*$ and every integer $r \geq 1$,*

$$\mu_{\text{irr}}(F((a/b)^r)) \leq \frac{1 - \theta}{\theta^* - \theta}.$$

Here $\mu_{\text{irr}}(\xi)$ is the supremum of the exponents ν for which $|\xi - p/q| < q^{-\nu}$ has infinitely many reduced rational solutions. In particular, $\mu_{\text{irr}}(F((31/4)^r)) < 301$ for every $r \geq 1$.

Proof. We follow the passage from linear forms to an exponent bound used by Zudilin at integer bases [4, p. 162]. Applying the estimates separately at each fixed base $(a/b)^r$ will give a bound independent of r . We use the polynomials U_n, V_n from Theorem 4.1, with $W_n = \deg U_n$. The additional estimate we need is a growth bound for their coefficients. In that construction the coefficient of F before cancellation is a sum of $O(n)$ Laurent monomials times two Gaussian binomial polynomials. Each Gaussian polynomial has nonnegative coefficients summing to at most 2^{27n+2} . Thus the sum of the absolute coefficients is $\exp(O(n))$. Its largest exponent is $K_n = (1091n^2 + 81n + 2)/2$, so its absolute value at a fixed $x > 1$ is at most $x^{K_n} \exp(O(n))$. The cyclotomic estimate in the same proof bounds the normalising multiplier by $x^{W_n - K_n} \exp(O_x(n))$. Multiplication gives

$$|U_n(x)| \leq x^{W_n} \exp(O_x(n)) \quad (x > 1 \text{ fixed}).$$

Set $\xi = F(a/b)$, $Q_n = b^{W_n} U_n(a/b)$ and $P_n = b^{W_n} V_n(a/b)$. With

$$\alpha = (C_1 - C_0) \log a, \quad \tau = C_0 \log a - C_1 \log b > 0,$$

we have $\log(Q_n \xi - P_n) = -\tau n^2 + o(n^2)$ and $|Q_n| \leq \exp(\alpha n^2 + o(n^2))$.

For a rational approximation with denominator q , we will choose a remainder at most $1/(2q)$ without letting $|Q_n|$ grow unnecessarily. The decay rate suggests taking n^2 near $(\log q)/\tau$. The reason for this choice is the following integer comparison. For integers A, B, p, q with $q > 0$, if $L = A\xi - B$ and $2q|L| \leq 1$, then

$$|L| \leq |A| |\xi - p/q|.$$

When $Ap - Bq = 0$ this is equality. Otherwise the nonzero integer $Ap - Bq$ gives $1/q \leq |L| + |A| |\xi - p/q|$, which proves the inequality. We fix $0 < \eta < \tau$ and use the preceding estimates in the form

$$e^{-(\tau+\eta)n^2} \leq Q_n \xi - P_n \leq e^{-(\tau-\eta)n^2}, \quad |Q_n| \leq e^{(\alpha+\eta)n^2}$$

for all sufficiently large n . The upper remainder bound determines our choice: for a sufficiently large denominator q , take $n = \lceil \sqrt{\log(2q)/(\tau - \eta)} \rceil$. Then $2q(Q_n \xi - P_n) \leq 1$. The preceding integer argument now applies to $(A, B) = (Q_n, P_n)$ for every numerator p . Also $Q_n \neq 0$: otherwise $Q_n \xi - P_n$ would be an integer strictly between 0 and 1. The lower remainder bound and the coefficient bound now give

$$|\xi - p/q| \geq e^{-(\alpha+\tau+2\eta)n^2} = q^{-(\alpha+\tau+2\eta)/(\tau-\eta)-o(1)},$$

since $n^2 = \log(2q)/(\tau - \eta) + O_\eta(\sqrt{\log q} + 1)$.

Letting $\eta \downarrow 0$, we deduce the bound $1 + \alpha/\tau = (1 - \theta)/(\theta^* - \theta)$. A common power multiplies α and τ by r , so the quotient is unchanged, although the constants in the approximation inequality may depend on r . For $31/4$, the exact interval bounds in the companion record, [Section 2.5](#), give $\theta < 0.4036982$ and $\theta^* > 0.40568$. The quotient increases with θ and decreases with θ^* in this range, so

$$\frac{1 - \theta}{\theta^* - \theta} < \frac{1 - 0.4036982}{0.40568 - 0.4036982} = \frac{2981509}{9909} < 301.$$

The same section gives the rational certificates for these bounds and sharper enclosures of the constants. $\square$

4.1 Comparison with earlier criteria

Bundschuh and Väänänen's Theorem 2 at $\alpha = -1$ [2, p. 177] gives irrationality for $\log b / \log a < \theta_{BV} := 1/2 - 1/\pi^2$. Their q is the base $a/b > 1$, not its reciprocal. Their logarithmic derivative is $L_q(z) = \sum_{j \geq 1} (q^j + z)^{-1}$, so the value at $z = \alpha = -1$ is exactly $F(a/b)$. The rational height is $h(q) = a$, and their parameter $\lambda = \log h(q) / \log q$ is $1/(1 - \log b / \log a)$, which gives the displayed cutoff. Since $\pi^2 < 10$, one has $\theta_{BV} < 2/5 < \log 4 / \log 31$, so $31/4$ lies outside that sufficient region. The two sufficient regions differ on $[\theta_{BV}, \theta^*)$. Zudilin also notes an extension to noninteger rational bases

$p = r/s$ for the generalized q -logarithm when $\log |r| > c \log |s|$, with $c > 0$ computable but unspecified [5, Sec. 2, p. 4]. For F , the specialisation above gives $c = \mu$, the exponent bound of [4, p. 162].

The positive-remainder estimates require $x > 1$, so they give no statement for negative bases. The series diverges at $x = 1$. The companion record also checks the parameters at $7/2$ in its section on Bundschuh and Väänänen’s height criterion. The analytic irrationality result used there is their cited theorem.

5 Further questions

An irrationality argument based on these determinants would also require a bound for their coefficient denominators. After clearing them, the nonzero determinants must still tend to zero; positivity and the growth of the moment weights alone give no such bound. For comparison, the [rational-value model in the companion paper](#) has the same formal orders, leading coefficients and type of fixed-base power correction, although its target value is rational. The model and its denominator calculation are retained there with ordinary proofs as unformalised remarks. The companion paper also gives an [ordinary argument for the ratio-limit extension](#) under $a_{k+1}/a_k \rightarrow \rho \in (0, q^{-1})$, without the polynomial shift bound in Theorem 2.1. That extension remains unformalised.

A different question concerns the polynomial coefficients of the approximants, rather than their remainders. Van Assche’s little- q -Legendre construction [12, §3] and the multiple-orthogonality construction of Postelmans and Van Assche [13, §2] make the normalisation important. In the 2016 family, the companion paper proves positivity of the first matrix of a coefficient pencil for sizes $N \leq 8$, with real, interlacing roots below $F(p)$ for $p > 1$. The shifted minors and coefficientwise positivity are supported by the stated finite polynomial computations. The [coefficient-moment section](#) gives the definitions and the precise ranges. Positivity for every N and convergence of the largest root remain open. The general moment criteria of Wang and Zhu [15] and Sokal and Walrad [17], Berg’s factorial-power example [16, Theorem 5.1], and the quadrature construction of Golub and Welsch [18] provide the relevant comparisons. Using a recurrence from another family would require a proof that it holds for these coefficients. The root-of-unity method of Krattenthaler, Rochev, Väänänen and Zudilin [14], for example, concerns a separate construction.

5.1 Congruences and divided forms

At $3/2$, a polynomial Q of degree at most W has integral homogenised value $H_W(Q) = 2^W Q(3/2)$. Congruences at powers of 2 and 3 give finite counting criteria for combinations of such values. The companion paper treats [integer rescaling](#) and [endpoint congruences and signed sums](#). It also treats the rank-one case and proves the quantitative finite-fibre estimate used below. Equal residues give a useful irrationality form only when the divided remainder is also nonzero and small. Smith normal form describes the image of the evaluated lattice [6, Theorems 2.3–2.4], but does not estimate these real remainders.

For example, suppose that M integral rows (A_j, B_j) have remainders $e_j = A_j F(3/2) - B_j$. Let D be a positive integer, and suppose that the coefficient-pair subset sums

$\sum_{j \in I} (A_j, B_j)$ take Q values in $(\mathbb{Z}/D\mathbb{Z})^2$. Put $T = \sum_j |e_j|$ and suppose that at most k subsets give any fixed remainder within a residue class. The least and greatest subset remainders are

$$\sum_{e_j < 0} e_j \quad \text{and} \quad \sum_{e_j > 0} e_j;$$

their difference is T . Thus T bounds the width of the range in each residue class as well. For each positive integer n , the inequality

$$2^M > Qk \left(\left\lfloor \frac{nT}{D} \right\rfloor + 1 \right) \quad (16)$$

gives two subsets with the same residue vector whose remainder values differ by an amount strictly between 0 and D/n in absolute value. Indeed, in each residue class divide the range of remainders into half-open intervals of width D/n , starting at its least value. At most $\lfloor nT/D \rfloor + 1$ intervals are needed, including a last interval when the greatest value is an endpoint. If each interval contained only one remainder value, it would contain at most k subsets, contrary to (16). Thus some interval contains two different remainder values; the half-open convention makes their separation strictly less than D/n . Subtracting the two subset sums and dividing their coefficients by D gives an integral linear form of nonzero absolute value less than $1/n$. The [quantitative lemma and its proof](#) state the finite-set version without assumptions about Lambert series.

The elementary integer-base method provides a different comparison. Vandehey [10] and Duverney and Tachiya [11] use integer-base expansions. At a noninteger rational base the [cleared-tail recurrence](#) has a growing denominator contribution. The [clearing and tail-recurrence calculations](#) in the companion paper include the [coordinatewise obstruction at 3/2](#). They make no assertion about other approximation families. At the function level, Bell and Smertnig [9, Theorem 1.3] exclude Mahler equations for the divisor generating series, while leaving the arithmetic of a single rational argument undecided.

A construction problem. The unrestricted request for primitive polynomial rows and a signed combination whose divided remainder lies strictly between 0 and $1/n$ is equivalent to the irrationality of the target. The companion paper retains the full statement and proves this equivalence in its [section on divided linear forms](#). A more specific question must prescribe the approximation family. For the 2004 family one can instead ask whether a parameter direction improves θ^* . The [all-base degree restriction](#) proves an upper bound of $1/2$ for the degree cutoff. Its hypotheses require one polynomial family, with common leading constants in its degree, coefficient-height and remainder estimates at every fixed real base $x > 1$; the error terms may depend on x . Merely having estimates at $3/2$ does not suffice.

Under those hypotheses, the [nondecay corollary](#) goes beyond the failure of a sufficient criterion: after denominator clearing, the undivided forms cannot tend to zero at any rational base a/b with $b < a < b^2$, including $3/2$, even without a quadratic degree limit. If that limit exists, their absolute values tend to infinity in this strict region. Division by additional base-dependent content and families outside these hypotheses

remain unclassified. Equality in the rational-base cutoff is also unresolved by the estimates of Section 4.

A Verification and reproducibility

The [evidence record](#) indexes the existing Lean proofs and recorded Comparator comparisons at the [source revision](#). Theorem 2.1 is the polynomial-bound specialisation of the [geometric-moment proof](#) cited in the companion paper; its Comparator association remains pending. The broader ratio-limit argument and rational-value model there have ordinary proofs only. This revision has had no fresh Lean or Comparator run and no independent review.

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The [verification record](#) gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded.

Data availability. The cited source revision contains the Lean sources, toolchain and library manifest. The companion paper supplies the longer algebraic arguments and the precise scope of the finite computations.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.1	Lean · Comparator	Theorem 2.1	Lean · Comparator pending
Proposition 3.1	Lean · Comparator	Theorem 3.2	Lean · Comparator
Theorem 4.1	Lean · Comparator	Corollary 4.2	Lean · Comparator
Corollary 4.3	Lean · Comparator		